

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 02



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Recap of Previous Lecture



- ✓ **Topic** Representation of graph
- ✓ **Topic** Simple graph, multi graph and pseudo graph
- ✓ **Topic** Maximum number of edges in a simple graph
- ✓ **Topic** Maximum number of simple graph with n vertices
- ✓ **Topic** Terminologies related to graphs

Topics to be Covered



✓
Topic

Different types of graphs





Topic : Types of graphs

- ✓ 1. Complete Graph (K_n)
- ✓ 2. Cycle Graph (C_n)
- ✓ 3. Wheel Graph (W_n)
- ✓ 4. Connected Graph
- ✓ 5. Cyclic Graph
- ✓ 6. Acyclic Graph
- ✓ 7. Tree
- ✓ 8. Bipartite Graph
- ✓ 9. Complete Bipartite Graph



Topic : Complete Graph

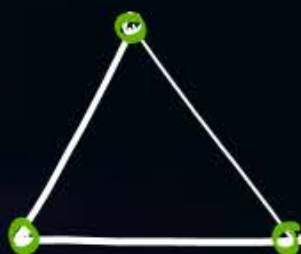
K_n

- A complete graph with n -vertices is denoted by ' K_n '
- K_n is a simple graph with n -vertices and all $n_2 = \frac{n(n-1)}{2}$ edges

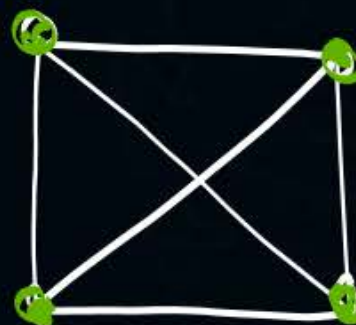
K_1



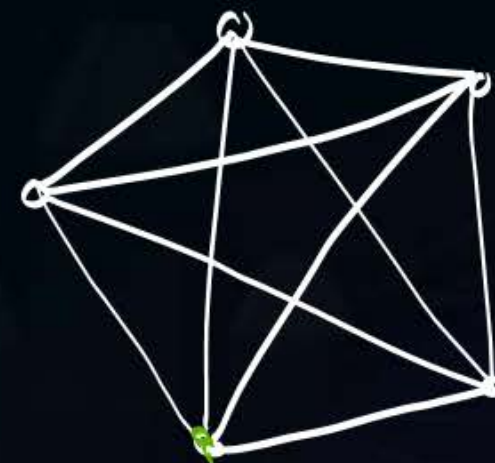
K_2



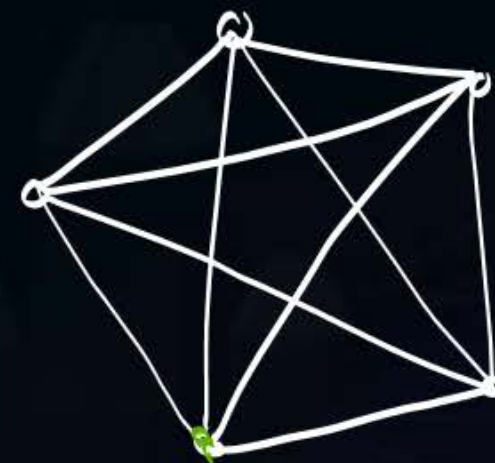
K_3



K_4



K_5



* In a complete graph K_n all n vertices are mutually adjacent

• In a complete graph K_n degree of each vertex is exactly ' $n-1$ ', $\therefore K_n$ is a ' $n-1$ ' regular graph

• In a complete graph K_n

$$|E(K_n)| = \frac{n(n-1)}{2}$$

No. of Edges in
Complete graph K_n



Topic : Cycle Graph



" C_n "

- ★ A simple graph with n -vertices and n -edges ($n \geq 3$) where all n edges form a cycle of length ' n ' is called a Cycle graph



$C_1 = \text{loop}$

- ★ A cycle graph with n -vertices is denoted by ' C_n '
- ★ Length of a cycle = No. of edges forming that cycle

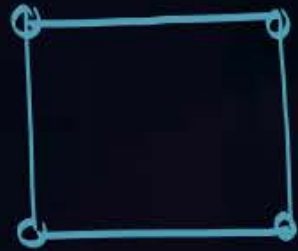


C_2

Two parallel edges



C_3

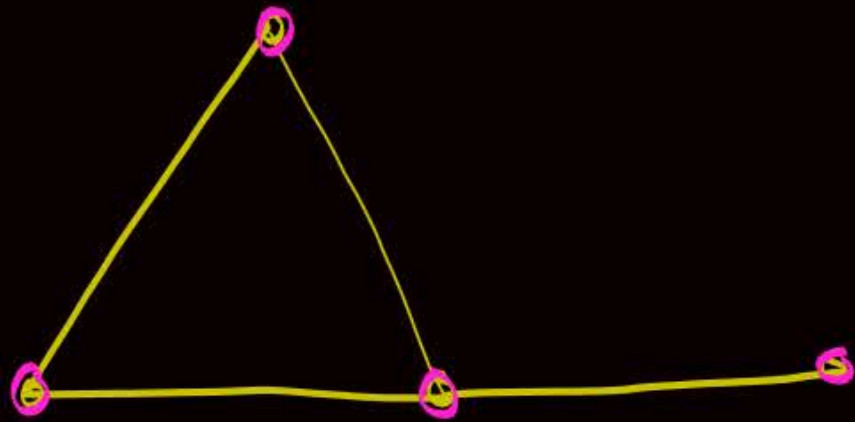


C_4

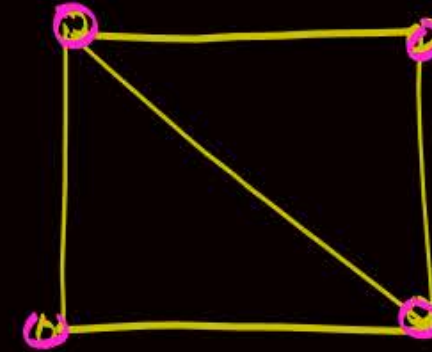


C_5

In a cycle graph C_n
 $|E(C_n)| = n$

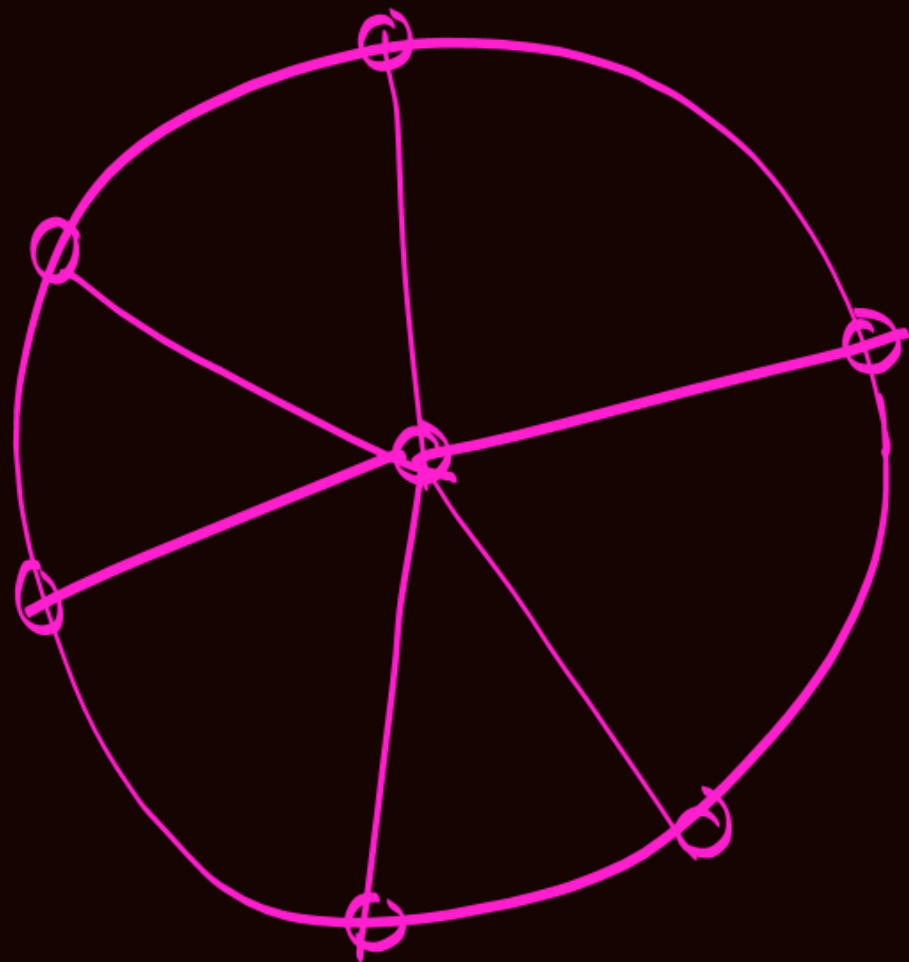


'4' vertices
and '4' edges
but is not a Cycle graph



Not a cycle graph
No. of vertices = 4
No. of edges = 5

W_n



$$W_n = C_{n-1} + 1\text{-hub}$$



Topic : Wheel Graph



' W_n '

A wheel graph with n -vertices is denoted by W_n .

A wheel graph with n -vertices can be obtained by connecting a hub vertex with all the vertices of cycle graph C_{n-1} ($W_n = C_{n-1} + 1\text{-hub}$)



$$W_4 = C_3 + 1\text{-hub}$$



$$W_5 = C_4 + 1\text{-hub}$$



$$W_6 = C_5 + 1\text{-hub}$$

In a wheel graph W_n

$$|E(W_n)| = (n-1) + (n-1) = 2(n-1) = (2n-2)$$

w.r.t.
 C_{n-1}

to connect
hub vertex with
(n-1) vertices of C_{n-1}

In a wheel graph W_n

- degree of hub vertex = $(n-1)$
- degree of Every other vertex = 3
(Except hub vertex)

Every wheel graph W_n
need not be a
regular graph

⊛ A wheel graph W_n is a regular graph
if and only if, $n-1 = 3$
i.e. $n=4$

The only wheel graph
which is regular as well
is " W_4 "

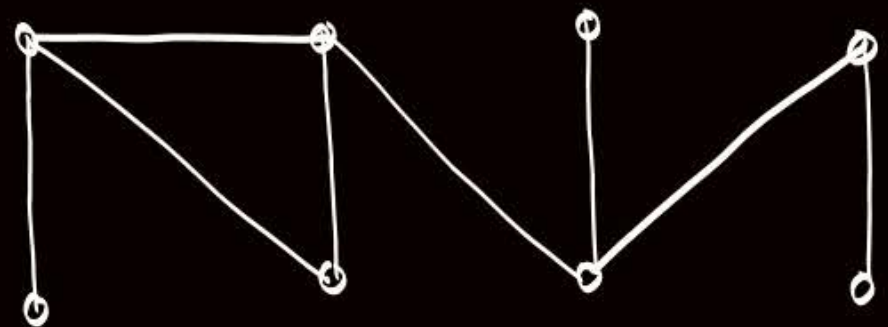


Topic : Connected Graph

→ A graph with two or more vertices is said to be connected if and only if there exist a path between every pair of vertices

★ A single isolated vertex may be treated differently in different scenarios.

eg:

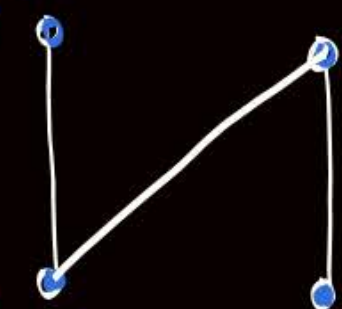
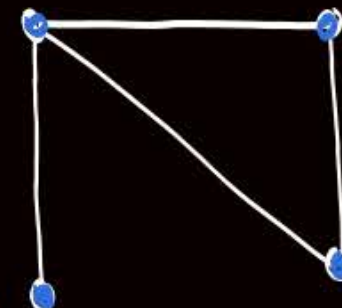


A single
Connected
Component

graph ' G_1 '

' G_1 ' is a connected graph + simple

it is
a Connected
Component



It is also
a Connected
Component

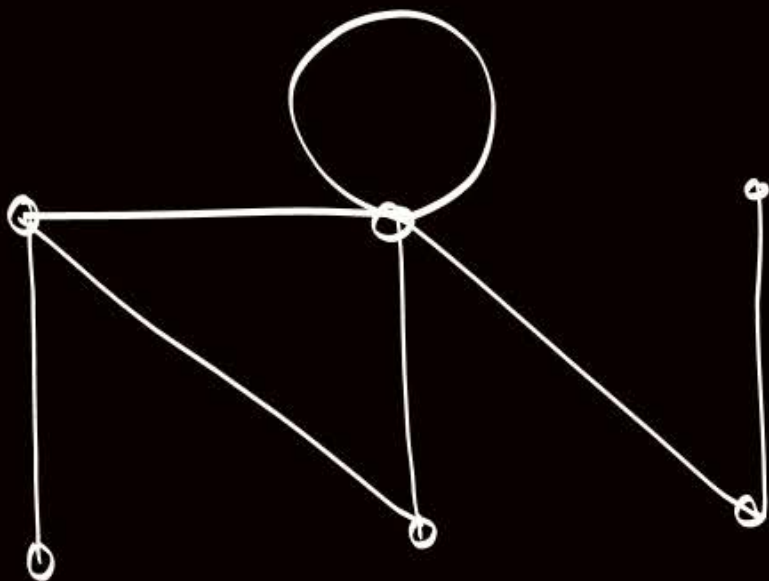
graph ' G_2 '

' G_2 ' is not a connected graph
(but simple)

In G_2 we have more than one
Connected Component

→ If graph is a Connected graph then there will be Exactly '1' Connected Component

→ If graph is a disconnected graph then it will have two or more Connected Components.



It is a Connected graph
but it is not a simple graph

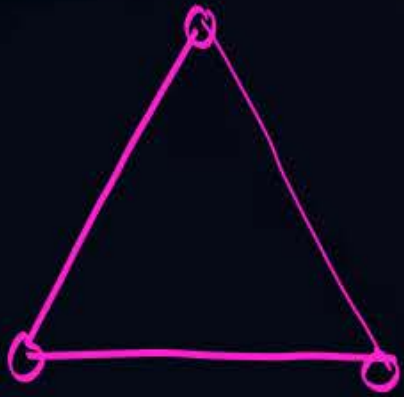
→ A Connected graph
may or may not
be a simple graph

→ K_n , C_n and W_n are always simple
as well as Connected graph

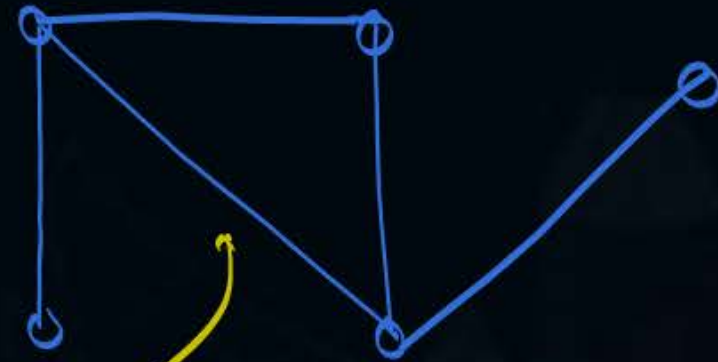


Topic : Cyclic Graph

- A graph with at least one cycle in it is called a cyclic graph

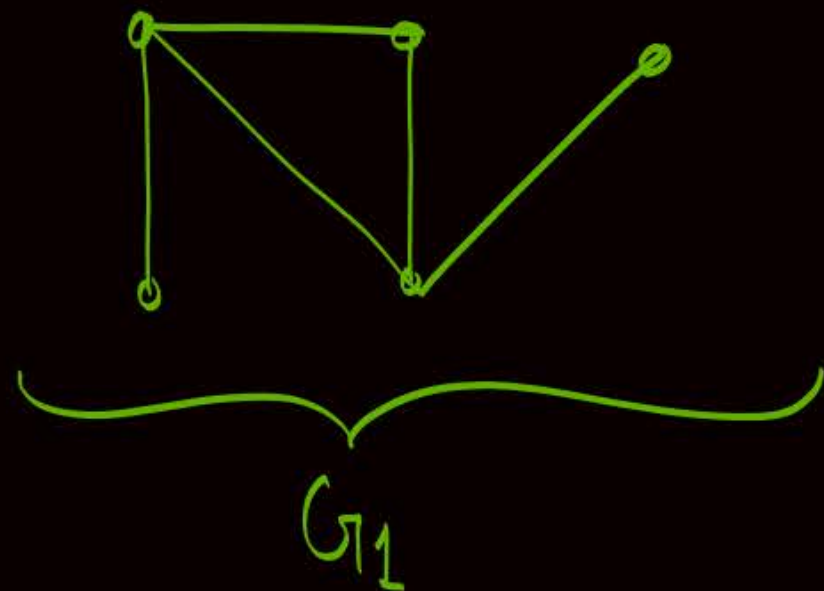


It is a cycle graph as well as a cyclic graph

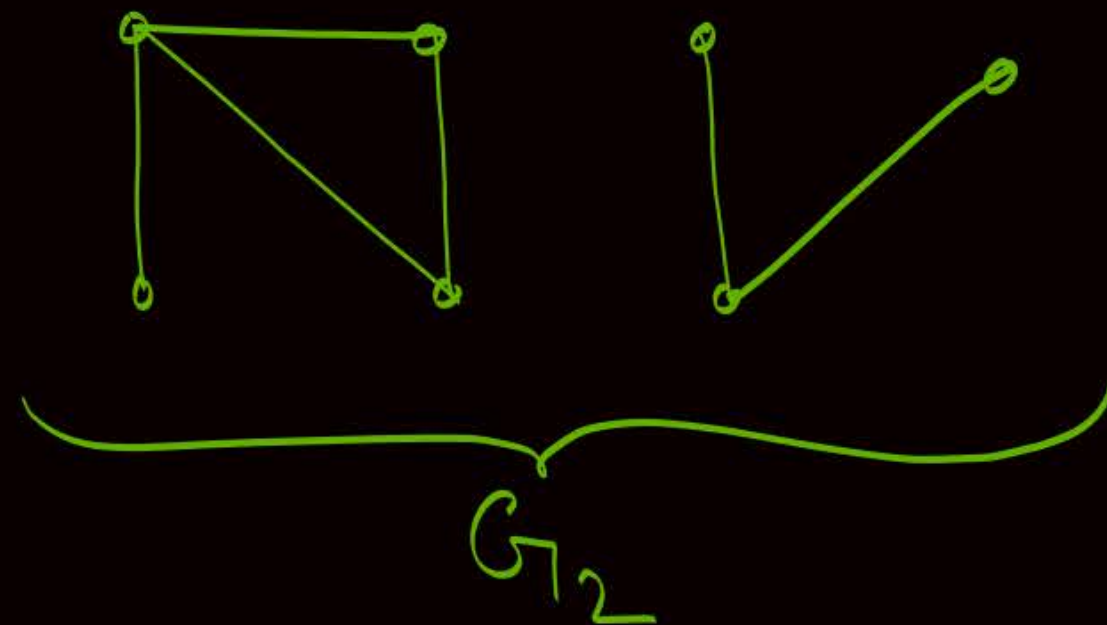


We have at least one cycle in it
∴ it is a cyclic graph,
but it is not a cycle graph

* Every cycle graph is also a cyclic graph,
but Every cyclic graph need not be a cycle graph



Cyclic + graph
+
Connected graph



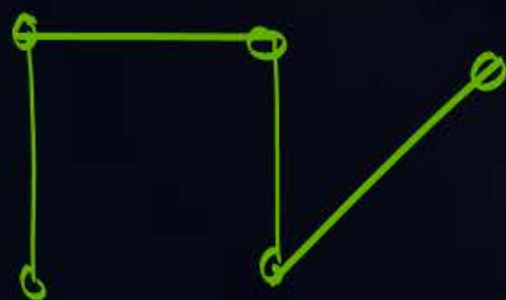
Cyclic + graph
+
disconnected graph

- * A cyclic graph may or may not be connected
- * A cyclic graph may or may not be simple



Topic : Acyclic Graph

→ A graph with no cycle in it is called an 'acyclic' graph.



Acyclic
+
Connected



Acyclic
+
Disconnected

Acyclic graph
are always
Simple



Topic : Tree



⊕ A tree is a connected and acyclic graph

⊕ A tree with n -vertices will have exactly ' $n-1$ ' edges

⊕ A collection of tree is called a forest

→ A forest is collection of 1 or more trees
→ A forest is a collection of one or more acyclic components

Q: A forest with n -vertices and $\underbrace{'K' \text{ Components}}_{\text{i.e. 'K' trees in the forest}}$ will have how many edges?

# Vertices	# trees	# Edges
n	1	$(n-1)$
n	2	$(n-1)-1 = n-2$
n	3	$(n-2)-1 = n-3$
$\vdots$	$\vdots$	$\vdots$
n	K	$(n-K)$

* In a simple graph with n -vertices, if number of edges are more than ' $n-1$ ' {i.e., $|E| > n-1$ }, then graph is guaranteed to be a cyclic graph {i.e., it can never be a tree}

* A graph with n -vertices and ' $n-1$ ' edges may or may not be a tree. { it may not be connected }
{ and may not be acyclic }

* A graph with n -vertices and less than ' $n-1$ ' edges can never be a connected graph { it may or may not be cyclic }

* In a graph with n -vertices if number of edges are more than $n-1$, {i.e., $|E| > n-1$ }, then graph is guaranteed to be cyclic, but may or may not be connected.



Topic : Bi-partite Graph

i.e., two vertices within the same set V_1 or V_2 must not be adjacent to each other

A bipartite graph (or bigraph) is a graph whose vertices can be partitioned into two sets V_1 and V_2 such that

every edge of the graph connects a vertex in set V_1 to a vertex in set V_2 .

*i.e. let $G=(V,E)$, then $V=\{V_1, V_2\}$ s.t. $V_1 \cap V_2 = \emptyset$
 $V_1 \cup V_2 = V$*

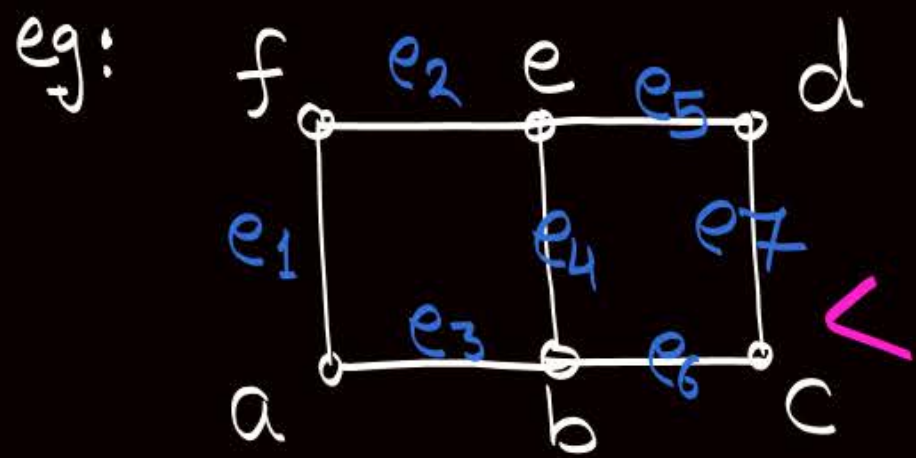
order triple

A bipartite graph can be represented using the notation $G=(V_1, V_2, E)$ where,

V_1 is the first set of the partition

V_2 is the second set of the partition and

E is the set of edges of graph G

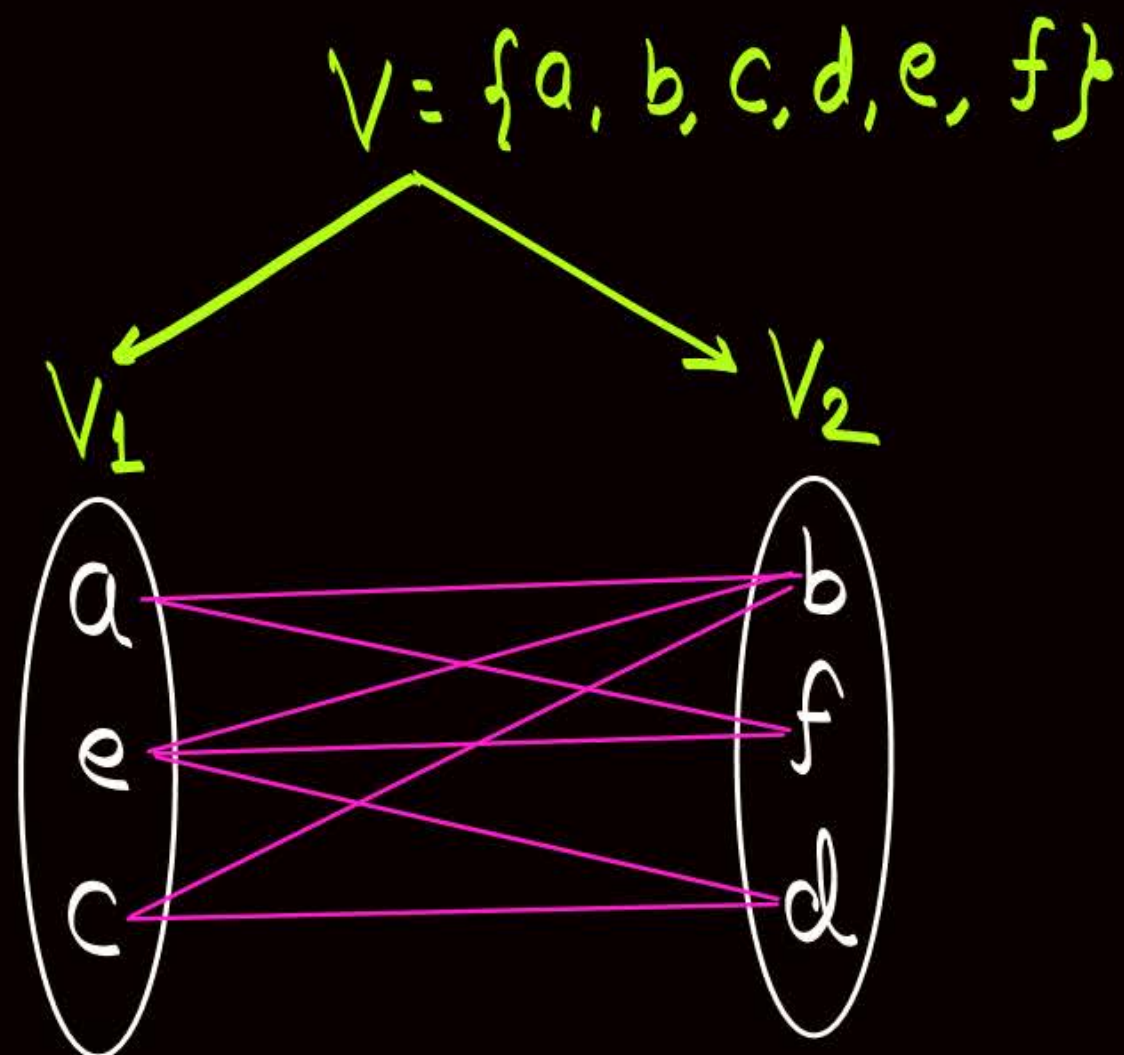


$$G = (V, E)$$

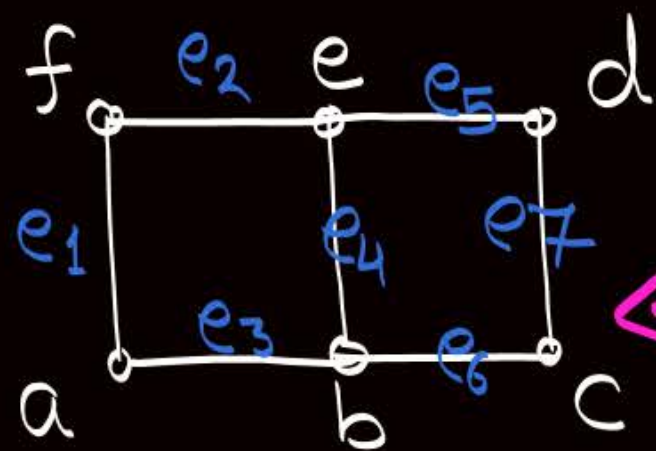
$$V = \{a, b, c, d, e, f\}$$

$$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$$

Check whether graph G is a bi-partite graph or not



eg:



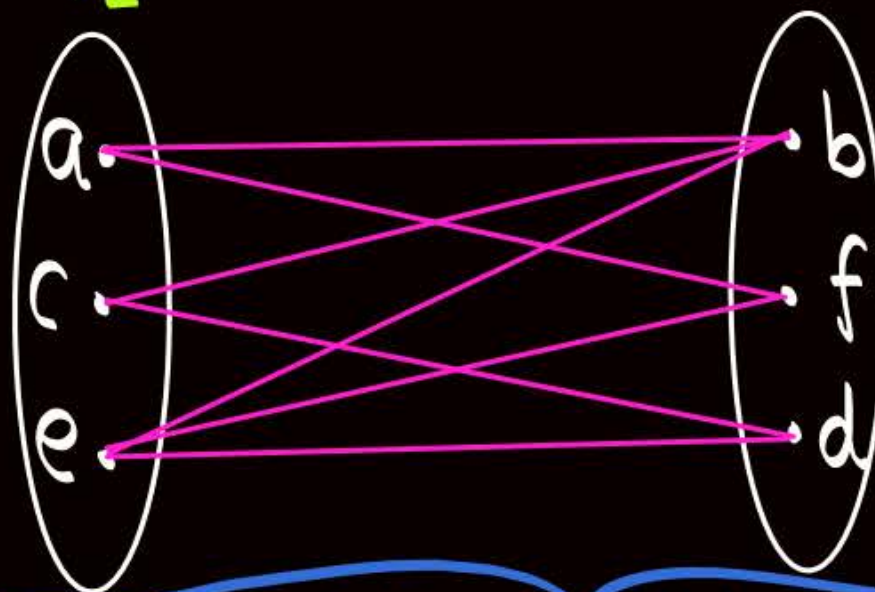
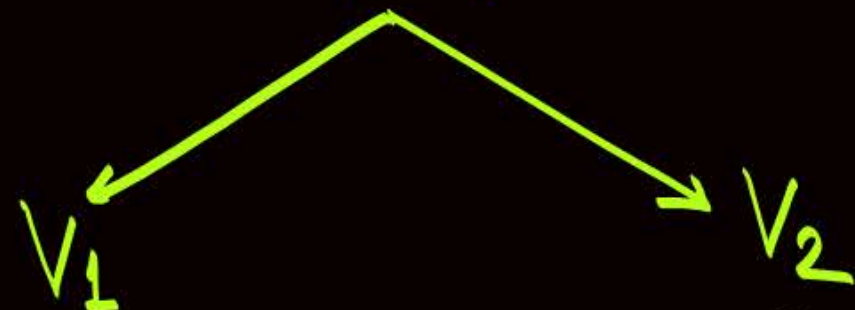
$$G = (V, E)$$

$$V = \{a, b, c, d, e, f\}$$

$$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$$

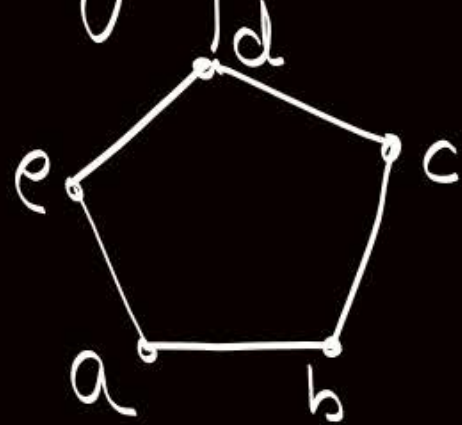
Check whether graph G is a bi-partite graph or not

$$V = \{a, b, c, d, e, f\}$$



We could partition the set of vertices into two subsets such that each edge of the graph connects a vertex of first subset to a vertex of another subset. i.e., No two vertices of same subset are adjacent to each other.
∴ Given graph is a bipartite graph

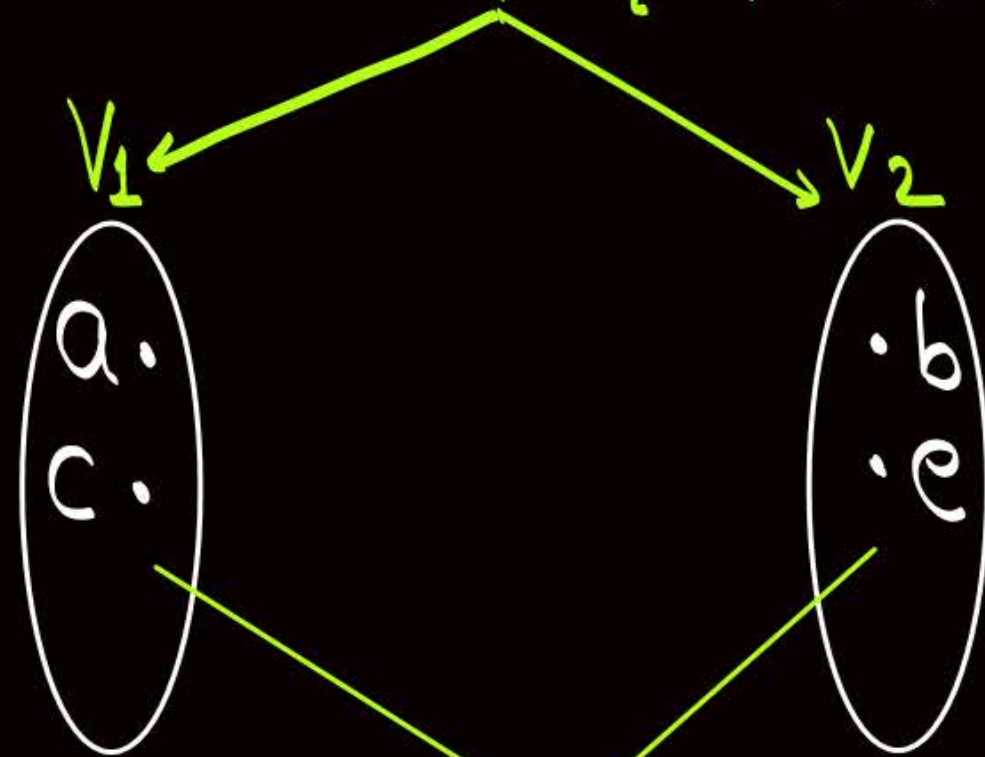
Q: Check whether graph or not.



$V = \{a, b, c, d, e\}$

the following graph is a bi-partite

$V = \{a, b, c, d, e\}$



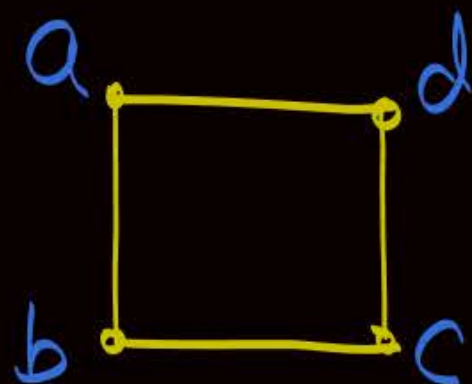
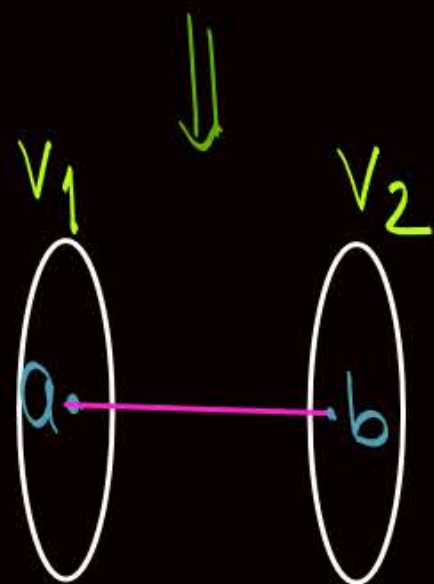
Vertex 'd' can not be placed in any subset

vertex 'd' can not be placed in subset V_1 because it is adjacent to vertex 'c'. Similarly vertex 'd' can not be placed in subset V_2 because it is adjacent to 'e'.

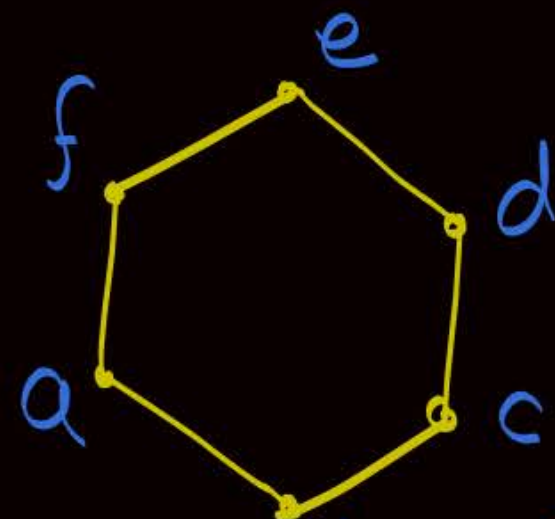
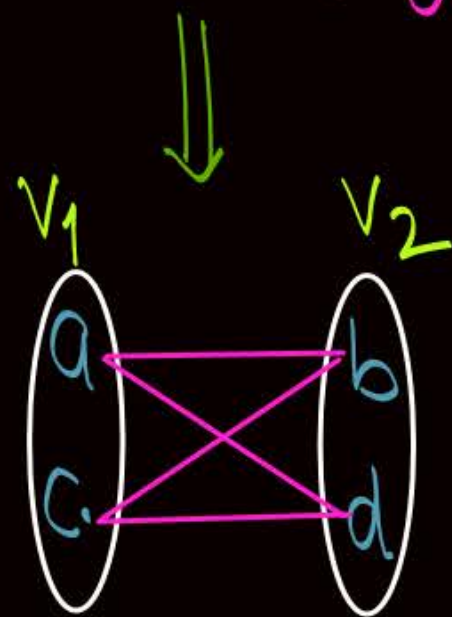
→ i.e. vertex set V can not be partitioned into two subsets.
As per given constraint, Graph is not a bipartite graph.



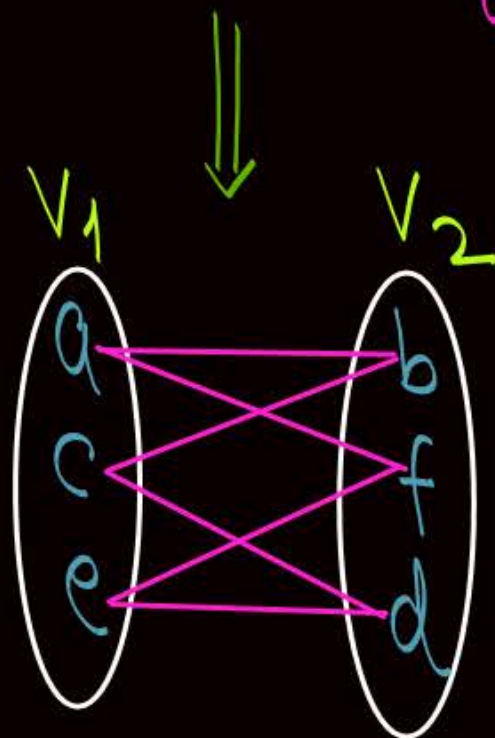
G_1 { cycle length = 0 }
 ↑
 Even



G_2 { cycle length = 4 }
 ↑
 Even



G_3 { cycle length = 6 }
 ↑
 Even



Note:- { ① In a graph if all cycles are of even length.
then that graph is always a bipartite graph

{ ② Tree is an acyclic graph { i.e. cycles of length = 0 }
∴ Every tree is a bipartite graph ↑
Even

{ ③ If there exist any cycle of odd length,
then graph can not be a bipartite graph

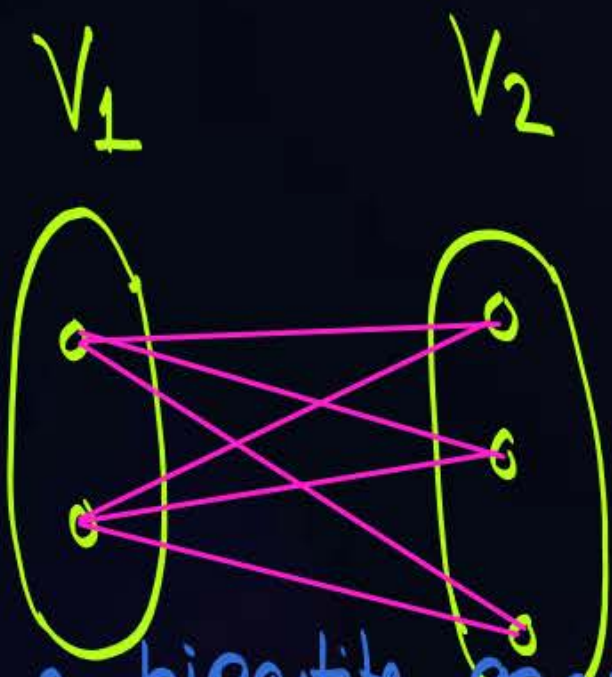
Note:

- ④ A graph with a single vertex can never be a bipartite graph
- ⑤ Every 'Null graph' with two or more vertices is a bipartite graph



Topic : Complete bipartite Graph

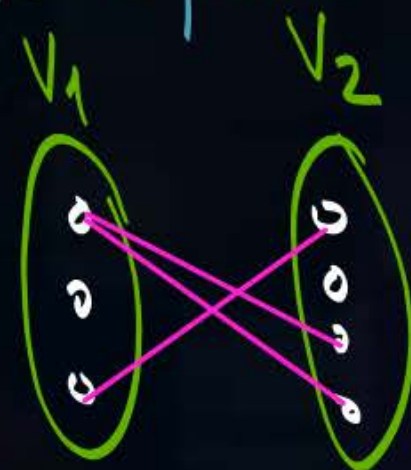
- A bipartite graph $G=(V_1, V_2, E)$ is called a complete bipartite graph if every vertex in set V_1 is adjacent to every vertex of set V_2 .



It is a bipartite graph,
And Every vertex of subset V_1 is adjacent
to Every vertex of subset V_2
or Complete bipartite graph

Every Complete bi-partite graph
is also a bipartite graph,
but Every bipartite graph need not
be Complete bipartite graph

eg.

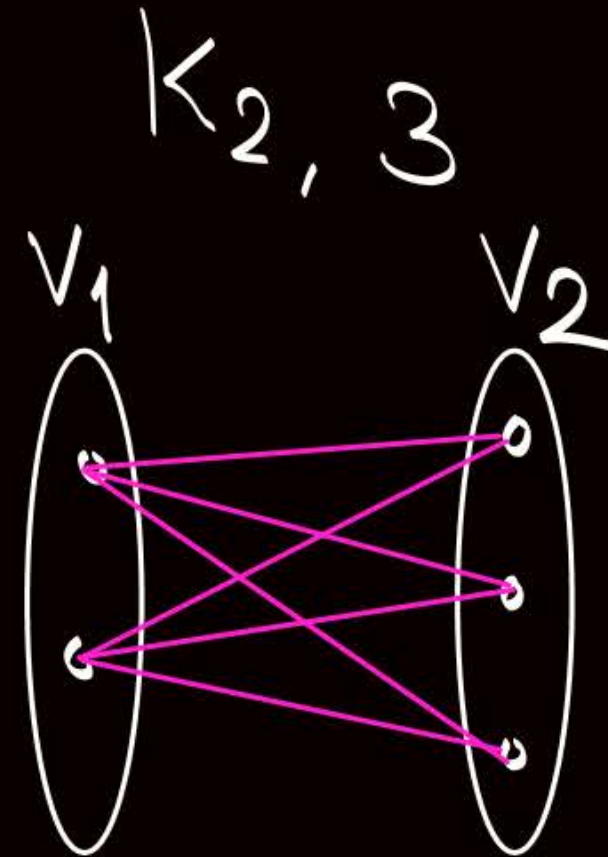
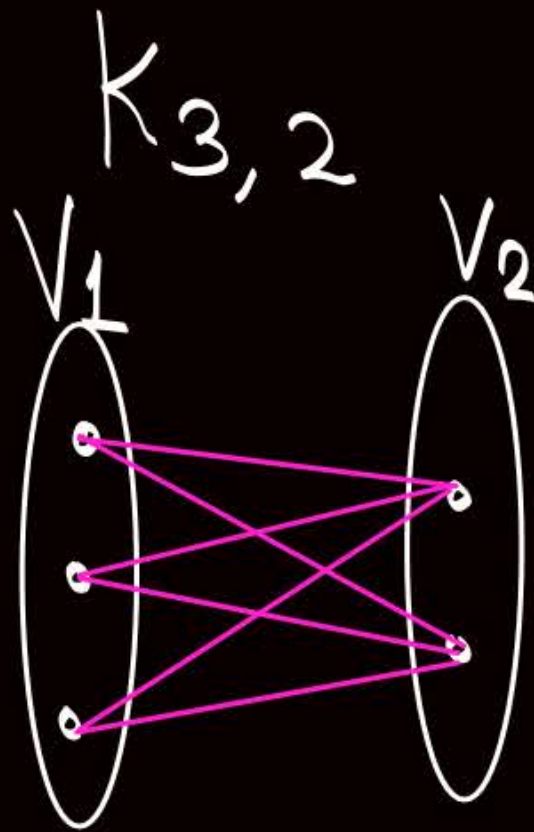


Bipartite ✓
Complete Bipartite ✗

★ A complete bipartite graph $G = (V_1, V_2, E)$

Where $|V_1| = m$ and $|V_2| = n$ is

represented by $K_{m,n}$

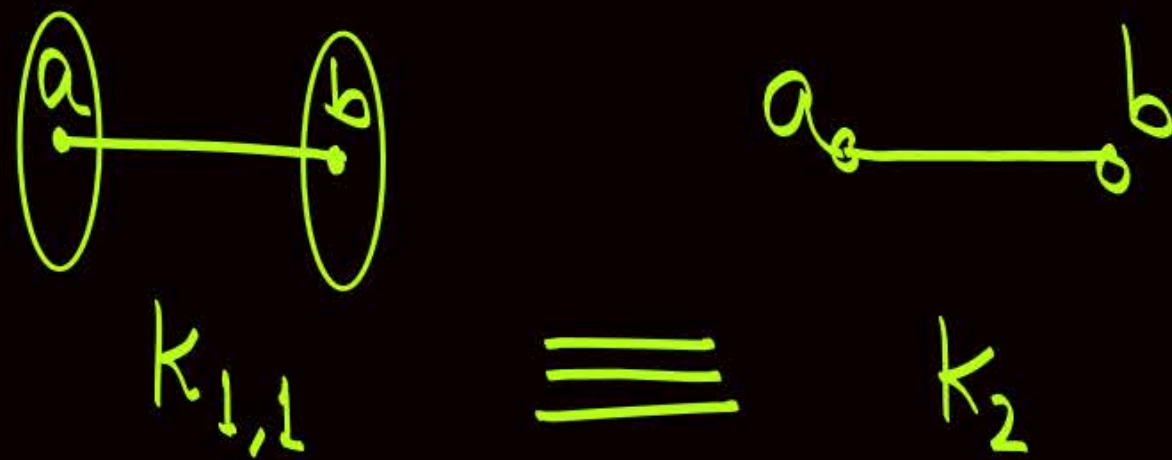


* In a Complete bipartite graph $K_{m,n}$

$$|E(K_{m,n})| = m \cdot n$$

In general, Complete bipartite graph is not
a Complete graph

{ A complete bipartite graph $K_{m,n}$
is a Complete graph if and only if ' $m=n=1$ '



Q:- Let G be a bipartite graph with ' n ' vertices, then maximum number of edges possible in graph G is?

$|V_1| + |V_2| = n$

Let $|V| = 10$ (Even)

V_1	V_2	Max. no. of Edges
1	9	9
2	8	16
3	7	21
4	6	24
5 ($n/2$)	5 ($n/2$)	25 = $\frac{n^2}{4}$
6	4	24
7	3	21
8	2	16
9	1	9

Let $|V| = 9$

V_1	V_2	Max. no. of Edges
1	8	8
2	7	14
3	6	18
4 ($\frac{n-1}{2}$)	5 ($\frac{n+1}{2}$)	20 = $(\frac{n-1}{2}) \cdot (\frac{n+1}{2})$
5 ($\frac{n+1}{2}$)	4 ($\frac{n-1}{2}$)	20 = $(\frac{n+1}{2}) \cdot (\frac{n-1}{2})$
6	3	18
7	2	14
8	1	8

Q:- Let G be a bipartite graph with ' n ' vertices,
then maximum number of edges possible in graph G is?

Maximum no. of Edges Possible

$\frac{n^2}{4}$ if $n = \text{Even}$

$\left(\frac{n+1}{2}\right)\left(\frac{n-1}{2}\right)$ if $n = \text{odd}$

Maximum no. of Edges Possible = $\left\lfloor \frac{n^2}{4} \right\rfloor$

K-partite graph:-

If we can partition the set of vertices into 'K' non-empty subsets such that no two vertices of same subset are adjacent to each other, then it is a K-partite graph.

→ If graph is a K -partite graph then graph is $(K+1)$ partite as well { Provided $(K+1)$ is less than or equal to the total number of vertices in the graph }

② If graph is a K -partite graph, then it need not be $(K-1)$ partite

Q.

Consider the set $S = \{1, w, w^2\}$, where w and w^2 are cube roots of unity. If $*$ denotes the multiplication operation, the structure $\{S, *\}$ forms



MCQ [2010: 1 Mark]

A

a group

B

a ring

C

an integral domain

D

a field

Q.

A binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is TRUE about $\oplus$?



MCQ [2013: 1 Mark]

- ☐ A Commutative but not associative
- ☐ B Both commutative and associative
- ☐ C Associative but not commutative
- ☐ D Neither commutative nor associative

Q.

Let G be a group with 15 elements. Let L be a subgroup of G . It is known that $L \neq G$ and that the size of L is at least 4. The size of L is_____.



NAT [2014 (Set-3): 1 Mark]

Q.

There are two elements x, y in a group $(G, *)$ such that every element in the group can be written as a product of some number of x 's and y 's in some order. It is known that

$$x*x = y*y = x*y*x*y = y*x*y*x = e$$

Where e is the identity element. The maximum number of elements in such a group is_____.

NAT [2014 (Set-3): 2 Marks]

Q.



Consider a set U of 23 different compounds in a Chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U . Consider the following statements:

- (I) Each compound in $U \setminus S$ reacts with an odd number of compounds.
- (II) At least one compound in $U \setminus S$ reacts with an odd number of compounds.
- (III) Each compound in $U \setminus S$ reacts with an even number of compounds.

Which one of the above statements is ALWAYS TRUE?

MCQ [2016 (Set-2): 2 Marks]

- | | |
|-------------------|------------------|
| A Only I | B Only II |
| C Only III | D None |

Q.

Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is ____.



NAT [2018: 1 Mark]

Q.



Let G be an arbitrary group.

Consider the following relations on G :

$R_1: \forall a, b \in G, aR_1b$ if and only if $\exists g \in G$ such that $a = g^{-1}bg$

$R_2: \forall a, b \in G, aR_2b$ if and only if $a = b^{-1}$

Which of the above is/are equivalence relation/relation(s)?

MCQ [2019: 1 Mark]

- A** R_1 and R_2
- B** R_1 only
- C** R_2 only
- D** Neither R_1 nor R_2

Q.

Let G be a group of 35 elements. Then the largest possible size of a subgroup of G other than G itself is _____. **NAT [2020: 1 Mark]**



Q.

Let G be a group of order 6, and H be a subgroup of G such that $1 < |H| < 6$. Which one of the following options is correct?



MCQ [2021: 2 Marks]

A

Both G and H are always cyclic.

B

G is always cyclic, but H may not be cyclic.

C

G may not be cyclic, but H is always cyclic.

D

Both G and H may not be cyclic.

Q.



Which of the following statements is/are TRUE
for a group G ?

MSQ [2022: 1 Mark]

If for all $x, y \in G$, $(xy)^2 = x^2y^2$, then G is commutative.

A

If for all $x \in G$, $x^2 = 1$, then G is commutative. Here, 1 is the
identity element of G .

B

If the order of G is 2, then G is commutative.

C

If G is commutative, then a subgroup of G need not be
commutative.

D

Q.

A pennant is a sequence of numbers, each number being 1 or 2. An n -pennant is a sequence of numbers with sum equal to n . For example, $(1,1,2)$ is a 4-pennant. The set of all possible 1-pennants is $\{(1)\}$, the set of all possible 2-pennants is $\{(2), (1,1)\}$ and the set of all 3-pennants is $\{(2,1), (1,1,1), (1,2)\}$.

Note that the pennant $(1,2)$ is not the same as the pennant $(2,1)$.

The number of 10-pennants is _____. **NAT [2014 (Set-1): 2 Marks]**





2 mins Summary



✓
Topic

Different types of graphs

Slide

THANK - YOU